CH 4 - Inner Product Spaces

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We defined abstract vector spaces to model $\mathbb{R}^n$.

So far, the focus has been on algebra. Geometry is rarely discussed.

In $\mathbb{R}^n$, we can talk about lengths, distance, and angles via DOT PRODUCT

Recall that in $\mathbb{R}^n$, $\vec{x} \cdot \vec{y} = \begin{bmatrix} x_1 \\ \dots \\ x_n \end{bmatrix} \cdot \begin{bmatrix} y_1 \\ \dots \\ y_n \end{bmatrix} = x_1 y_1 + \dots + x_n y_n$

Observations:

1. $\vec{x} \cdot \vec{x} = x_1^2 + \dots + x_n^2 = \|\vec{x}\|^2$
2. $\text{dist}(x, y) = \|\vec{y} - \vec{x}\| = \sqrt{(\vec{x} - \vec{y}) \cdot (\vec{x} - \vec{y})}$
3. $\vec{x} \cdot \vec{y} = \|x\|\|y\| \cos\theta \implies \theta = \arccos \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\|\|\vec{y}\|}$

We can view dot product as $f : V \times V \to \mathbb{R}$, $f(x, y) = x \cdot y \in \mathbb{R}$ that satisfies

1. $f(x, y) = f(y, x)$, symmetry
2. $f(cx + z, y) = cf(x, y) + f(z, y)$, linearity in the first argument(second argument is same by symmetry)
3. $f(x, x) \geq 0 \ \forall x$, $f(x, x) = 0 \iff x = 0$, positive-definite
4. $\frac{f(x,y)}{\|x\|\|y\|} \in [-1, 1]$, can be obtained from the above 3 properties.

> 📘 **Definition — Inner Product**
>
> Let V be a vector space over $\mathbb{F}$. An **inner product** on V is a function
>
> $$\langle\ ,\ \rangle : V \times V \to \mathbb{F}$$
>
> satisfying:
>
> 1. Symmetry: $\langle \vec{x}, \vec{y} \rangle = \overline{\langle \vec{y}, \vec{x} \rangle}$
> 2. Linearity in arguments: $\langle a\vec{x} + y, \vec{z} \rangle = a\langle \vec{x}, \vec{z} \rangle + \langle \vec{y}, \vec{z} \rangle$
> 3. Positive-definite: $\langle \vec{x}, \vec{x} \rangle \geq 0$ and $\langle \vec{x}, \vec{x} \rangle = 0 \iff \vec{x} = \vec{0}$

Examples

1. $V = \mathbb{R}^n$, $\langle \vec{x}, \vec{y} \rangle = \vec{x} \cdot \vec{y}$ is an inner product
2. $V = P_2(\mathbb{R})$, $\langle a_0 + a_1 x + a_2 x^2, b_0 + b_1 x + b_2 x^2 \rangle = \begin{bmatrix} a_0 \\ a_1 \\ a_2 \end{bmatrix} \cdot \begin{bmatrix} b_0 \\ b_1 \\ b_2 \end{bmatrix} = a_0 b_0 + a_1 b_1 + a_2 b_2$
3. $[L^2 \text{ inner product}] V = P_n(\mathbb{R})$

   $\langle p(x), q(x) \rangle = \int_0^1 p(x) q(x)\, \mathrm{d}x$

   Claim: the above is an inner product on $P_{n(\mathbb{R})}$

   1. Symmetry: $\langle p, q \rangle = \int_0^1 p(x) q(x)\, \mathrm{d}x = \int_0^1 q(x) p)(x)\, \mathrm{d}x = \langle q, p \rangle$

2. Linearity $\langle ap(x)+q(x), r(x)\rangle = \int_0^1 (ap(x)+q(x))r(x)\,\mathrm{d}x = a\int_0^1 p(x)r(x)\,\mathrm{d}x + \int_0^1 q(x)r(x)\,\mathrm{d}x = a\langle p(x), r(x)\rangle + \langle q(x), r(x)\rangle$
3. Absolute-definite $\langle p(x), p(x)\rangle = \int_0^1 p(x)p(x)\,\mathrm{d}x = \int_0^1 p^2(x)\,\mathrm{d}x \geq 0$

   If $\langle p(x), p(x)\rangle = 0$, then $\int_0^1 p^2(x)\,\mathrm{d}x = 0 \Longrightarrow p(x) = 0$

4. $V = M_{2\times 2}(\mathbb{R})$

   $\langle \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \begin{bmatrix} x & y \\ z & w \end{bmatrix} \rangle = ax + by + cz + dw$

   Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, B = \begin{bmatrix} x & y \\ z & w \end{bmatrix}$

   $\langle A, B\rangle = \mathrm{tr}(B^T A)$ (Frobenius Inner Product on $M_{n\times n}(\mathbb{R})$)

   Special case, $M_{n\times 1} = \mathbb{R}^n$

   $\langle \vec{x}, \vec{y}\rangle_F = \mathrm{tr}(\vec{y}^T\vec{x}) = \mathrm{tr}([x_1y_1 + ... + x_ny_n]) = x_1y_1 + ... + x_ny_n = \vec{x}\cdot\vec{y}$

Propositions of Inner Products

1. By symmetry, we have linearity in second argument in $\mathbb{R}^n$: $\langle \vec{x}, a\vec{y} + \vec{z}\rangle = a\langle \vec{x}, \vec{y}\rangle + \langle \vec{x}, \vec{z}\rangle$
2. $\langle \vec{x}, \vec{0}\rangle = \langle \vec{0}, \vec{v}\rangle = 0$
3. Every finite-dimensional vector space V has an inner product
4. Every non-zero vector space V has infinitely many inner product.
5. $V = \mathbb{C}^n, \langle x, y\rangle = x_1\cdot\overline{y}_1 + ... + x_n\cdot\overline{y}_n$
6. The second arguments are conjugate linear $\langle \vec{x}, a\vec{y} + \vec{z}\rangle = \overline{a}\langle \vec{x}, \vec{y}\rangle = +\langle \vec{x}, \vec{z}\rangle$

Example:

Intuitively, $\langle \vec{x}, \vec{x}\rangle = \begin{bmatrix} i \\ 0 \\ ... \\ 0 \end{bmatrix} \cdot \begin{bmatrix} i \\ 0 \\ ... \\ 0 \end{bmatrix} = i^2 + 0^2 + ... + 0^2 = -1$

We define $\langle \begin{bmatrix} z_1 \\ ... \\ z_n \end{bmatrix}, \begin{bmatrix} w_1 \\ ... \\ w_n \end{bmatrix} \rangle = z_1\overline{w}_1 + z_2\overline{w}_2 + ... + z_n\overline{w}_n$

Then, $\langle \vec{x}, \vec{x}\rangle = \begin{bmatrix} i \\ 0 \\ ... \\ 0 \end{bmatrix} \cdot \begin{bmatrix} i \\ 0 \\ ... \\ 0 \end{bmatrix} = i\cdot\overline{i} + ... + 0\cdot\overline{0} = -(-1) = 1$ satisfies absolute-definite

Proof

2. $\langle \vec{x}, \vec{0}\rangle = \langle \vec{x}, \vec{0} + \vec{0}\rangle = \langle \vec{x}, \vec{0}\rangle + \langle \vec{x}, \vec{0}\rangle - \langle \vec{x}, \vec{0}\rangle = \langle \vec{x}, \vec{0}\rangle - \langle \vec{x}, \vec{0}\rangle = 0$
3. $V \cong \mathbb{R}^n$

   $\langle x, y\rangle = [x]_B \cdot [y]_B$ is an inner product
4. There exists an inner product $\langle , \rangle$

   Consider $c\langle , \rangle,\ c \in \mathbb{R}_{>0}$

   There are infinitely many inner product

Recall:

1. $\vec{x} \cdot \vec{x} = x_1^2 + \ldots + x_n^2 = \|\vec{x}\|^2$
2. $\text{dist}(x, y) = \|\vec{y} - \vec{x}\| = \sqrt{(\vec{x} - \vec{y}) \cdot (\vec{x} - \vec{y})}$
3. $\vec{x} \cdot \vec{y} = \|x\|\|y\| \cos\theta \Longrightarrow \theta = \arccos \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\|\|\vec{y}\|}$

Definition — Norms, Distance, Angles

Let V be an inner product space(a vector space with a fixed choice of inner product)

The **norm** of $\vec{x} \in V$ is $\|\vec{x}\| = \sqrt{\langle \vec{x}, \vec{x} \rangle}$ is a non-negative real number

The **distance** between $\vec{x}, \vec{y} \in V$ is $\text{dist}(\vec{x}, \vec{y}) = \sqrt{\langle \vec{x}, \vec{y} \rangle}$ is non-negative real number

IMPORTANT: DO NOT FORGET THE SQUARE ROOT

Examples:

1. If $V = \mathbb{R}^n$, $\langle , \rangle$ be dot product. These definitions are the familiar ones from Euclidean Geomtery.
2. $V = M_{2\times 2}(\mathbb{R})$, $\langle , \rangle$ be the Frobenius inner product
   $\| \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \| = \sqrt{\langle A, A \rangle} = \sqrt{1^2 + 2^2 + 3^2 + 4^2} = \sqrt{30}$
3. $V = P_2(\mathbb{R})$, $\langle p, q \rangle = \int_0^1 p(x)q(x)\,\mathrm{d}x$
   $\text{dist}(1 - x^2, 2 + 3x + 2x^2) = \sqrt{1 - x^2 - 2 - 3x - 2x^2} = \sqrt{-1 - 3x - 3x^2}$
   $= \sqrt{\langle -1 - 3x - 3x^2, -1 - 3x - 3x^2 \rangle} = \sqrt{\int_0^1 (-1 - 3x - 3x^2)^2\,\mathrm{d}x} = \sqrt{\frac{153}{10}}$

Theorem — Properties of Norm

Let V be an inner product space

1. $\|\vec{x}\| \geq 0$ and $\|\vec{x}\| = 0 \iff \vec{x} = \vec{0}$ (Positive-definite)
2. $\|c\vec{x}\| = |c|\ \|\vec{x}\|$ (Homogeneity)
3. $\|\vec{x} + \vec{y}\| \leq \|\vec{x}\| + \|\vec{y}\|$ (Triangle Inequality)

Proof

2. $\|c\vec{x}\| = \sqrt{\langle c\vec{x}, c\vec{x} \rangle} = \sqrt{c\bar{c}\langle \vec{x}, \vec{x} \rangle} = \sqrt{|c|^2\ \langle \vec{x}, \vec{x} \rangle} = |c|\ \sqrt{\langle \vec{x}, \vec{x} \rangle} = |c|\ \|\vec{x}\|, \quad c \in \mathbb{F}$
3. $$\begin{aligned} \|\vec{x} + \vec{y}\|^2 &= \langle \vec{x} + \vec{y}, \vec{x} + \vec{y} \rangle = \langle \vec{x}, \vec{x} \rangle + \langle \vec{x}, \vec{y} \rangle + \langle \vec{y}, \vec{x} \rangle + \langle \vec{y}, \vec{y} \rangle \\ &= \|\vec{x}\|^2 + \langle \vec{x}, \vec{y} \rangle + \overline{\langle \vec{x}, \vec{y} \rangle} + \|\vec{y}\|^2 \\ &= \|\vec{x}\|^2 + 2\ \text{Re}(\langle \vec{x}, \vec{y} \rangle) + \|\vec{y}\|^2 \\ &\leq \|\vec{x}\|^2 + 2|\langle \vec{x}, \vec{y} \rangle| + \|\vec{y}\|^2 = \|\vec{x}\|^2 + 2\|\vec{x}\|\ \|\vec{y}\| + \|\vec{y}\|^2 = (\|\vec{x}\| + \|\vec{y}\|)^2 \end{aligned}$$

   $\Longrightarrow \|\vec{x} + \vec{y}\| \leq \|\vec{x} + \vec{y}\|$

□

Theorem — Cauchy-Schwarz Inequality

$$|\langle x, y\rangle| \leq \|x\| \ \|y\|$$

Basic idea: if $\langle , \rangle$ is dot product on $\mathbb{R}^n \leq 1$

$|\langle x, y\rangle| = |x \cdot y| = \|x\| \ \|y\| \ |\cos\theta| \leq \|x\| \ \|y\|$

Definition —

If $x, y \in V$ are non-zero, let the angle between them to be $\theta = \arccos\left(\frac{\langle x,y\rangle}{\|x\|\|y\|}\right)$

$$\frac{\langle x, y\rangle}{\|x\|\|y\|} \leq 1$$

by Cauchy-Schwarz

Question: when $\theta = 90$?

Ans: $\frac{\langle x,y\rangle}{\|x\|\|y\|} = \cos 90 = 0 \implies \langle x, y\rangle = 0$

Definition — Orthogonal in Inner Product Space

Let V be an inner product space. We say $\vec{x}, \vec{y} \in V$ are orthogonal

$$\vec{x} \perp \vec{y} \iff \langle \vec{x}, \vec{y}\rangle = 0$$

Examples:

1. $V \in \mathbb{R}^2, \langle , \rangle =$ dot product. $e_1 \perp e_2 \iff \langle \begin{bmatrix}1\\0\end{bmatrix}, \begin{bmatrix}0\\1\end{bmatrix}\rangle = 0$
2. $V = P_2(\mathbb{R}), \langle p, q\rangle = \int_0^1 p(x)q(x)\,\mathrm{d}x$

   $p = 1, q = x$ are not orthogonal b/c $\langle p, q\rangle = \langle 1, x\rangle = \int_0^1 x\,\mathrm{d}x = \frac{1}{2} \neq 0$

   If we define $\langle p, q\rangle = \int_{-1}^1 pq\,\mathrm{d}x$

   Then $1 \perp x \iff \langle 1, x\rangle = \int_{-1}^1 x\,\mathrm{d}x = 0$

Theorem — Pythagorean Thm

Let V be an inner product space.

If

$$x \perp y \implies \|\vec{x} + \vec{y}\|^2 = \|\vec{x}\|^2 + \|\vec{y}\|^2$$